

Implementation of a customized content management system for digital marketing based on LG webOS Signage and beacons usage

Arturo Andrade
Escuela de Ingeniería de
Sistemas y Computación
Universidad Peruana de
Ciencias Aplicadas
Lima, Perú
u201323940@upc.edu.pe

Hugo Bullon
Escuela de Ingeniería de
Sistemas y Computación
Universidad Peruana de
Ciencias Aplicadas
Lima, Perú
u201410268@upc.edu.pe

Alfredo Barrientos
Escuela de Ingeniería de
Sistemas y Computación
Universidad Peruana de
Ciencias Aplicadas
Lima, Perú
pcsiabar@upc.edu.pe

Miguel Cuadros
Escuela de Ingeniería de
Sistemas y Computación
Universidad Peruana de
Ciencias Aplicadas
Lima, Perú
pcsimcua@upc.edu.pe

Abstract— Due to the way marketing has evolved over time, creating innovative and powerful messages is insufficient. Today, the companies focus on the communication with each customer in a unique way considering its specific preferences, interests, and individual needs. In this context, a variety of digital signage devices is appearing in the market. This type of product has technologies that are not always fully exploited by their manufacturers and end users. This is the case of the LG Digital Signage devices, which are integrated with a beacon (Bluetooth Low Energy technology-BLE) that is not harnessed by the different content management systems available on the market. On this basis, the research team proposes a content management system which can control the information shown in various LG Digital Signage devices. In addition, to customize the experience, the available beacon in the devices was used to create an interaction between the smartphones of the users and the LG Digital Signage monitor, avoiding direct contact with objects that could be manipulated by others. Consuming the information of an external recommendation engine the necessary information can be obtained. In the experiment, many users were asked to try the system to watch general content as well as the customized one in the available monitors. The effectiveness of the system was also evaluated.

Keywords—CMS, beacon, digital signage, digital marketing, technology, society, customer satisfaction, customized information

I. INTRODUCTION

Over the years, the companies have started to build awareness and value the information they hold and can get from their users. This is possible through digital media and other various ways to get it (Data Mining, clickstream, etc.). This information is crucial against an increasingly demanding public that for a long time was saturated with information not needed or not of their preference. Therefore, it is recognized that information customization often represents a great differentiation from traditional media of competition, a better brand recall and high chance of purchasing or return to the establishments [1].

The digital signage has achieved a special prominence in the marketing world being the direct replacement of the traditional billboards. Digital media allows to display all sorts of content through screens such as LCD, plasma and LED monitors or projectors. Given its popularity, new requirements made possible to add capabilities such as internet connection, cameras, Wi-Fi, Bluetooth, Beacons, etc. to this type of devices.

To control the information to be presented in different digital signage monitors, systems able to manage options such as the device management, the installation of the devices, the interface design, the assignments, etc. arose. This type of systems is known as CMS (Content Management System). However, the large majority focus on managing general information and refrain from exploiting the capacities that the devices are gaining. For instance, LG Digital Signage incorporated with Beacons are rarely exploited.

A Beacon is a transmitter device used to send a BLE signal (Bluetooth Low Energy) that uniquely identifies it. This signal can be detected by compatible devices that enters its radio range. The Beacon technology has different protocols: iBeacon created by Apple, the open-source Eddystone led by Google and AltBeacon created by Radius Networks [3]. These devices are mostly used for micro-location and provide information in general. Today, mobile devices and other gadgets with Bluetooth 4.0 support BLE [4].

Therefore, it is proposed as a solution the implementation of a complete and adaptable web content management system, capable of providing general and customized information using an external recommendation system and controlling the interactivity with the users due to the Beacon incorporated into the LG Digital Signage devices. For that purpose, the additional development of two mobile libraries (both for iOS and Android) is proposed. These libraries will be able to manage the reception of BLE signals and the sending of the user identifier near to the monitor, providing a better customized and interactive experience.

II. RELATED WORK

A. Content Management System for Digital Signage

There are many solutions where content management systems have been developed such as [5], which presented a content management system for audience and environment adaptive digital signage. In this solution, multiple contexts were hierarchically modelled for their adaptation. In addition, the content for several contexts was systematically stored and retrieved for its adaptation in runtime promoting its reuse.

In [6], a profitable management system of digital signage of autonomous style for Raspberry Pi signaling devices was developed. Between the management system and de signage device a SSH was used for communication. This system had two modules. The first one is the administrator module with web interface to manage the database with information of devices, authentication and contents. And the second one is

the media support module, which reproduces the content and transfers the screen image data to the target device.

A digital signage software solution, called Xibo4u, which provides customized information is presented in [7]. The open-source base software used to develop this solution is Xibo. Xibo4u is composed by an advertising server and content management category, digital screens that show advertising based on the client's specific categories and a smartphone app that allows the clients to designate their favorite advertising categories via Wi-Fi.

B. Data customization with Beacons

In [8], a new targeted advertising method is proposed. It is experimentally adapted to the digital signage systems by using an iPhone which acts as BLE beacons and Raspberry Pi 3 as a server connected to a signage. The iPhone user acts as transmitter of information and the module of script for scanning in the server converts this data into a JSON file constantly read by the same server, thus managing to customize the information shown.

Based on the research [9], the authors developed a system to provide information to the foreign tourist using the user's native language. The multilingual information service system combining the smartphone and the digital signage based on the iBeacon technology. This system had three functions such as showing multilingual notifications, having access to the multilingual web information, and changing the multilingual digital signage. The assessment experiment was conducted in the tourism area of Shirakawa-go in Japan, obtaining a good reception from users.

In [10], a solution called TAR was presented, which is a system using existing surveillance cameras and ubiquitous BLE signals to identify and track with precision the customers of a retail store allowing the targeted advertising assignment. In TAR, a single-camera tracking algorithm was designed. It traced people with precision and then TAR was extended to a multiple camera setting to identify people using the installed cameras. With TAR, the vicinity information was seized thanks to the BLE of the smartphones, the movement patterns of the cross camera, and the single-camera tracking algorithm to achieve high tracking precision and the identification of many people.

III. SYSTEM OVERVIEW

On the basis of the research carried out on the different projects of digital signage, the features to be developed in order to enhance the user experience with the system and the contents to be presented in the digital signage devices were established. The development of the project was divided in two stages: design and implementation.

A. Design

In this stage, the work environments and platforms to be used were defined. In the case of webOS Signage [8] and the web application, the code editor Visual Studio Code was used. To develop the web services, Visual Studio 2019 was used as development environment. XCode 11 was used to develop the iOS library and Android Studio for the Android library. Subsequently, the way the content would be displayed in the device was defined. The usage of iBeacon as the standard to be used in the devices due to its compatibility with iOS and Android was defined.

Finally, the features of the content management system were defined: Contents, Devices, Recommendation engines, Assignments and Libraries. In addition, it was defined that the system would be set by companies and each of them would have users able to interact with the web application of the system.

B. Implementation

After defining all the features to be developed in the different applications of the project, the design of the logical and physical architecture of the project started, followed by its development.

1) *Services*: The web app, the webOS app and the mobile libraries use web services for integration. These services were developed using the REST architecture, the C# programming language and .NET Framework 4.5 in Visual Studio 2019, later deployed in Azure. The services connect to a SQL database which stores all the information of the content management system. Two web services were created: Web Service - CMS and Web Service - Libraries. The first one allows the interaction between the content management system and the webOS application. The second one allows the interaction between the content management system and the mobile libraries. The complete interaction can be seen in Fig.1.

Within Web Service – CMS, the HTML of the contents that will be shown in the LG Digital Signage device are generated. The contents stored in the database used a specific template that has its parameters specified in an array of JSON objects with the following structure:

```
{ "name": parameter_name, "type": parameter_type }
```

Each JSON object has stored the information submitted from the CMS web application (text, images, or colors).

The Web Service – Libraries assigns the user customized information in the content management system so that it can be later shown in the device using a JSON object as content.

2) *WebOS application*: To develop the webOS application, the base project webOS Signage SDK 4.0 was used. The animations and the requests to REST services were developed using JQuery 1.12.4. The application enables the configuration of a company by means of a 4-character code allowing the existence of 436 companies within the system. The usage of SCAP API was added in order to control the internal Beacon of the LG Digital Signage device, assigning the respective Beacon UUID to the configured company. The application requests the contents that must be shown in the device (assigned, customized or empty content) to the Web Service – CMS.

3) *Web application*: To develop the web application, ReactJS was used as front-end framework and UIKit as style sheet. Components were developed for each of the features (Contents, Devices, Recommendation engines, Assignments and Libraries). Dynamic forms were generated for the different templates within the system. React Router was used to manage the different features and accelerate the development process.

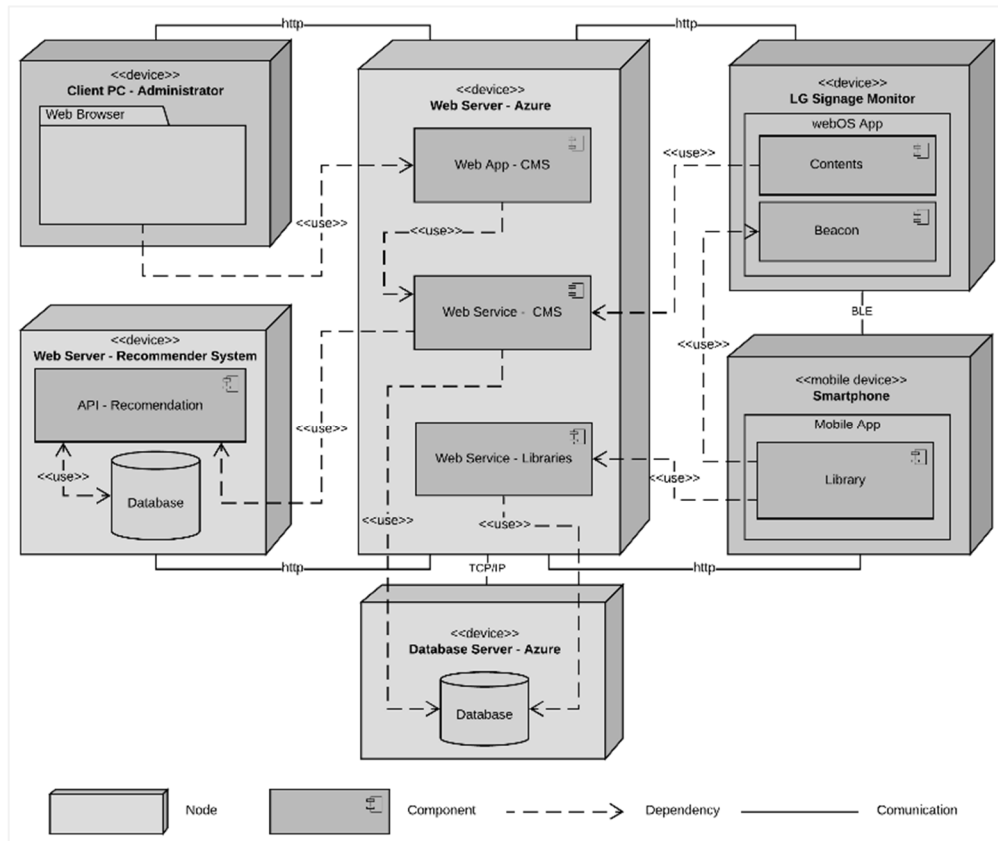


Fig. 1. Complete system interaction. Logical Architecture.

The feature “Assignment” uses the contents generated in “Contents” as well as a device and a recommendation engine. It is possible to set an assignment by time ranges and activate the use of Beacon of the device to show the customized information. It is necessary to assign a recommendation engine and a template to the information that will be shown with this feature. The interface can be seen in Fig. 2.

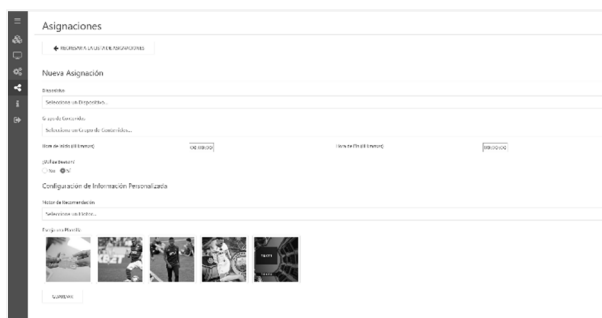


Fig. 2. User Interface of web application. Assignment feature.

4) *Mobile libraries:* To develop the iOS mobile library, Core Location and Core Bluetooth were used. Functions for the detection of beacons and calls to Web Service – Libraries were implemented, which include the list of available beacons and notifications of proximity to them. The library only searches and detects the beacons defined in the system. It does not detect or interact with external beacons. It was defined that 45 seconds after detecting a beacon in the system the calling to deactivate the customized information will be made. The library was improved so that a development team

only uses 3 lines in its implementation. The function only needs a key and a user identifier to be used.

To develop the Android mobile library, Java was used as programming language. The Retrofit library was used for the calls to the REST services and the AltBeacon library, for the beacon detection. The beacons detected are compared with the list of beacons available in the system. It was defined that 45 seconds after detecting a beacon in the system the calling to deactivate the customized information will be made. The resulting function only needs a key, a user identifier and the context of the application in development.

IV. VALIDATION

When the development of the project started, it was established that the application of the LG Digital Signage device would provide a high user experience underpinned by the designs to be displayed in the device. A simulation where a car dealership used the system to show advertising of different brands was created. This advertising was designed with the available templates in the system. The survey was conducted to 30 people between the ages of 16 and 26 who were contextualized in the car dealership simulation. They were asked to provide their personal information in the mobile application so that the system can obtain customized information from the recommendation engine developed for the simulation. At the end of the simulation, they were asked to complete a survey of 12 questions. The first question was about the mobile operating system and the other 11 questions had a 1 to 10 rating system. The results of the 11 rating questions are shown in Table I.

TABLE I
QUESTIONS AND AVERAGES OBTAINED FROM THE SURVEY

Questions	Average
Did you find friendly the advertisements shown in the LG Digital Signage device?	9.23
Did you find the QR code useful in the advertisements shown?	7.67
Do you consider that the design of the advertisements was friendly?	9.37
Was your experience with the mobile application friendly?	8.90
Was it pertinent that the device advertisement was customized with your name?	9.53
Did you like the customized messages?	9.10
Would you download a mobile application of a store/shopping mall?	8.47
Would you download a mobile application so that the store/shopping mall can show you customized advertisements?	8.37
How would you rate the mobile interaction with the LG Digital Signage devices?	8.97
Would you prefer this kind of advertisements against the static advertisements?	9.83
Do you find important, as a user, to receive customized information according to your preferences?	9.83

In addition, 53.3% had an Android device and the remaining 46.7%, an iOS device. It can be observed that the results obtained in each question were highly positive. There was a question where 70% of the survey respondents rated with a 10, how pertinent was that the device advertisement was customized with their name. Finally, more than 65% rated with an 8 or more the interaction created between their mobile device and the LG Digital Signage device.

V. CONCLUSIONS

The previous conducted research guided the correct definition of the features needed by a content management system which uses a beacon to show customized content.

The system development focused on the ease of use for the user (a company, a store, a shopping mall), adding libraries of easy implementation so that the user can develop its mobile application with connectivity to the available LG Digital Signage devices.

To prove that the contents generated by the content management system provided a highly user experience, a survey was conducted to a sample of 30 people. Out of the 11 questions, a mean of 9.02 was obtained from a 1 to 10 score range, demonstrating that the system implemented had a very high acceptance among the public and that more than 70% considers as pertinent the advertising customization and that it provides added value to digital marketing.

All procedures performed in studies involving human participants were in accordance with the ethical standards of the institutional and/or national research committee and with the 1964 Helsinki declaration and its later amendments or comparable ethical standards.

VI. REFERENCES

[1] CLIENTO, Marketing digital. "Importancia de la personalización en marketing digital", <https://blog.cliento.mx/personalizacion-marketing-digital>, last accessed 2020/11/09.

[2] K2B, "The Beacon War - Know what is a iBeacon, a Eddystone and a AltBeacon", <https://www.k2bindia.com/ibeacon-vs-eddystone-vs-altbeacon>, last accessed 2020/11/09.

[3] UOC (Universitat Oberta de Catalunya), "La tecnología beacons: una revolución en alza para la experiencia de usuario y las estrategias de marketing", <https://www.uoc.edu/portal/es/news/actualitat/2016/099-beacons.html>, last accessed 2020/11/09.

[4] Lee, W., Lee, H.-W., Lee, Y.-T., Ryu, W., & Choi, M, "Content management system for environment adaptive digital signage", International Conference on Information and Communication Technology Convergence (ICTC). doi:10.1109/ictc.2014.6983201, 2014.

[5] Mishima, K., Sakurada, T., & Hagiwara, Y., "Low-cost managed digital signage system with signage device using small-sized and low-cost information device", 14th IEEE Annual Consumer Communications & Networking Conference (CCNC). 10.1109/ccnc.2017.7983172, 2017.

[6] Lee, W. & Kim, N. & Yoon, E. & Jin, J. & Lee, M., Xibo4u: Digital Signage for Personalized Advertisement. Journal of the Korea Institute of Information and Communication Engineering. 19, pp. 1359-1367. 10.6109/jkiice.2015.19.6.1359, 2015.

[7] Okamoto, T., Sasaki, Y., & Suzuki, S., Targeted advertising using BLE beacon, IEEE 6th Global Conference on Consumer Electronics (GCCE). 10.1109/gcce.2017.8229404, 2017.

[8] Ogi, T., Ito, K., & Ukegawa, S., Multilingual Information Service Based on Combination of Smartphone and Digital Signage. Advances in Network-Based Information Systems Lecture Notes on Data Engineering and Communications Technologies, pp. 644-653, 2017.

[9] Liu, X., Jiang, Y., Jain, P., & Kim, K.-H., TAR. Proceedings of the 16th Annual International Conference on Mobile Systems, Applications, and Services - MobiSys '18. 10.1145/3210240.3210342, 2018.

[10] WebOS Signage developer, "Discover webOS Signage", <http://webossignage.developer.lge.com/discover/webos-signage/discover-webos-signage>, last accessed 2020/11/09.